

- Pipeline Workflow, Columns, Sources, and Business Rules
 - Workflow Diagram
 - Source Workflows
 - CMETS Source
 - Effectiveness Source
 - JCC Source
 - Bay Allocation Source
 - Combined Workflow
 - Source-by-Source Workflow
 - 1. CMETS Source Workflow
 - 2. Effectiveness Source Workflow
 - 3. JCC Source Workflow
 - 4. Bay Allocation Source Workflow
 - Universal Matching Priority
 - Cross-Source Matching Map
 - CMETS Columns
 - Effectiveness Columns
 - Effectiveness Mapping Rules
 - Effectiveness-Derived Mapped Columns
 - JCC Columns
 - JCC Extraction Rules
 - JCC Mapping and Computed Columns
 - JCC COD, TGNA, and GNA Logic
 - Bay Allocation Columns
 - Full PDF Table Columns
 - Bay Allocation Excel Columns
 - Bay Mapping Columns
 - Bay Number Selection Logic
 - Exact Row Identification and Source Dependency
 - Exact Column Lineage for Bay Mapping
 - Bay Allocation PDF Columns Extracted
 - Bay Allocation Fallback Row Match in Detail

Pipeline Workflow, Columns, Sources, and Business Rules

This document explains the current workflow in non-technical language. It is based on the code in the CMETS, Effectiveness, JCC, Bay Allocation, and mapping handlers.

Workflow Diagram

These diagrams use only plain text, arrows, and simple string characters so they can be copied into any Markdown or text viewer.

Source Workflows

CMETS Source

```
CMETS PDFs
-> Read agenda/minutes PDFs
-> Classify meeting as GNA or LTA from first pages
-> Reject non-application pages/tables
    (GNARE, bulk consumer, drawee tables)
-> Extract application rows
-> Keep rows with at least one ID:
```

```
GNA/ST II Application ID
LTA Application ID
Application ID under Enhancement 5.2 or revision
-> Normalize IDs, dates, status, type, state, voltage, BESS, PSP
-> Deduplicate application rows
-> 01_cmets_extracted.xlsx
```

Effectiveness Source

```
Effectiveness PDFs
-> Read PDF pages
-> Extract application_id, applicant, project type, MW break-up
-> Extract substation, state, region, expected_date
-> Normalize IDs, dates, capacity values
-> Build lookup:
    application_id -> effectiveness record
-> 02_effectiveness_extracted.xlsx
```

JCC Source

```
JCC PDFs
-> Page gate:
    Pooling + Quantum + Connectivity
-> Use LLM extraction for exact raw table-column text
-> Extract only these raw JCC columns:
    Pooling Station
    Connectivity Applicant
    Connectivity Quantum (MW)
    Under Grantee scope Gen Commissioning /Connectivity line schedule
    Connectivity Start Date under GNA and Connectivity Effectiveness date
-> Compute total COD and COD_Found from the Generation section only
-> Extract effective_date from the connectivity effectiveness date text
-> If COD_Found is True, decide GNA/TGNA from effective_date versus current run date
-> Keep JCC rows for Layer 4 matching
-> 04_jcc_extracted.xlsx
```

Bay Allocation Source

```
Bay Allocation PDFs
-> Page gate:
    Name of Substation + RE Capacity Granted + Margin on Existing
-> Detect bay allocation table headers
-> Extract name_of_substation
-> Extract substation_coordinates
-> Extract region
-> Extract 220kV bay records:
    220kV Bay No. + 220kV Name of Entity
-> Extract 400kV bay records:
    400kV Bay No. + 400kV Name of Entity
-> Build lookup:
    voltage bucket + entity name -> bay record
-> 05_bayallocation_extracted.xlsx
```

Combined Workflow

```
01_cmets_extracted.xlsx
-> Module 3: CMETS + Effectiveness mapping
-> Match CMETS IDs against effectiveness application_id
    order: GNA/ST II Application ID -> LTA Application ID -> 5.2 ID
-> Update applicant, substation, state, quantum, region, type, capacity columns
-> Update GNA Operationalization Date from expected_date when later
-> Update Additional Capacity Date using GNA -> LTA -> 5.2
-> 03_cmets_effectiveness_mapped.xlsx
-> Layer 4: CMETS-first JCC mapping
-> Match CMETS IDs inside JCC connectivity_applicant
    order: GNA/ST II Application ID -> LTA Application ID -> 5.2 ID
-> Once JCC row is identified:
    extract Bay No from schedule_current_jcc_ists_scope if present
-> From matched JCC row:
```

```

total COD = sum MW values under Generation only when each value has a date
            and COD/Commissioned keyword
COD_Found = True when such Generation COD/Commissioned entries are found
effective_date = date from Connectivity Start Date under GNA and
                Connectivity Effectiveness date
if COD_Found is True and current run date is after effective_date:
    GNA = total COD
if COD_Found is True and effective_date is after current run date:
    TGNA = total COD
-> 04_cmets_jcc_mapped.xlsx
-> Module 6: Bay mapping
-> Check Bay No (JCC)
    -> if present:
        use JCC Bay No and skip Bay Allocation search for that row
    -> if not present:
        search Bay Allocation lookup using:
            CMETS Voltage level + CMETS Name of Developers
-> Populate Bay No, Substation Name, Coordinates, Bay No Source
-> 06_cmets_bay_mapped.xlsx

```

Source-by-Source Workflow

1. CMETS Source Workflow

1. Read CMETS PDFs.
2. Extract meeting metadata from first pages. If GNA keyword count is comparable to or higher than LTA, write meeting number/date to **CMETS GNA Approved** and **CMETS GNA Meeting Date**; otherwise write to LTA columns.
3. Reject pages/tables that are not connectivity application rows, especially GNARE, bulk consumer, and drawee tables.
4. Extract application rows with the CMETS prompt.
5. Keep only rows that have at least one primary ID: **GNA/ST II Application ID**, **LTA Application ID**, or **Application ID under Enhancement 5.2 or revision**.
6. Normalize IDs, dates, status, project type, state, voltage, battery, and PSP columns.
7. Deduplicate by **GNA/ST II Application ID**, **LTA Application ID**, **Project Location**, and **Name of Developers**.

2. Effectiveness Source Workflow

1. Read effectiveness PDFs.
2. Extract each row's **application_id** plus applicant, region, project type, capacity values, substation, state, and **expected_date**.
3. Build a lookup keyed by **application_id**.
4. This source is not matched by developer name. It is matched only by IDs from CMETS against effectiveness **application_id**.

3. JCC Source Workflow

1. Read JCC PDFs.
2. Accept only pages that contain **Pooling**, **Quantum**, and **Connectivity**.
3. Use the LLM to extract the exact raw JCC table text. Do not infer, normalize, or logically derive the raw values during extraction.
4. Extract only these raw JCC columns from the table: **Pooling Station**, **Connectivity Applicant**, **Connectivity Quantum (MW)**, **Under Grantee scope Gen Commissioning /Connectivity line schedule**, and **Connectivity Start Date under GNA and Connectivity Effectiveness date**.
5. Add computed columns after raw extraction: **total COD**, **COD_Found**, **effective_date**, **TGNA**, and **GNA**.

- 6. `total COD` and `COD_Found` are derived only from the `Generation:` part of `Under Grantee scope Gen Commissioning /Connectivity line schedule`.
- 7. `effective_date` is extracted from `Connectivity Start Date` under `GNA and Connectivity Effectiveness date`.
- 8. JCC rows are not mapped immediately during extraction. They are used later by the CMETS-first Layer 4 mapping.
- 9. TGNA/GNA calculation happens only after a CMETS row is matched to a JCC row.

4. Bay Allocation Source Workflow

- 1. Read Bay Allocation PDFs.
- 2. Accept pages that contain substation name, RE capacity granted, and margin-on-existing markers.
- 3. Extract substation name, coordinates, region, 220kV bay records, and 400kV bay records.
- 4. Build a lookup with two buckets: `220kv` and `400kv`.
- 5. Each lookup entry stores Bay Allocation `entity_name`, `bay_no`, `name_of_substation`, and `substation_coordinates`.

Universal Matching Priority

Most cross-source matching follows this business order:

- 1. Use `GNA/ST II Application ID`.
- 2. If not found, use `LTA Application ID`.
- 3. If not found, use `Application ID under Enhancement 5.2 or revision`.

Effectiveness and JCC depend heavily on this ID cascade. Bay Allocation fallback does not use application IDs; it uses voltage plus developer/entity name after the JCC bay-number check has failed.

Cross-Source Matching Map

Workflow step	CMETS column used	Other source column searched	Match order / rule	Columns u after match
CMETS + Effectiveness merge	<code>GNA/ST II Application ID</code> , <code>LTA Application ID</code> , <code>Application ID under Enhancement 5.2 or revision</code>	Effectiveness <code>application_id</code> lookup key	Try GNA first, then LTA, then 5.2. The first ID found in the lookup wins.	Name of Developer Substation Application (Quantum II), Region Project, i capacity bre columns
GNA Operationalization Date update	<code>GNA/ST II Application ID</code> , then <code>LTA Application ID</code>	Effectiveness <code>application_id</code> ; date value from <code>expected_date</code>	Try GNA, then LTA. If <code>expected_date</code> is later than CMETS <code>GNA Operationalization Date</code> , update CMETS date.	<code>GNA Operationalization Date</code> , <code>GNA Operationalization (Yes/No)</code>
Additional Capacity Date update	<code>GNA/ST II Application ID</code> , <code>LTA Application ID</code> , <code>Application ID</code>	Effectiveness <code>application_id</code> ; date value from <code>expected_date</code>	Try GNA, then LTA, then 5.2. If <code>expected_date</code> is later than CMETS date, update CMETS date.	<code>Date from additional capacity added</code>

Workflow step	CMETS column used	Other source column searched	Match order / rule	Columns used after match
	ID under Enhancement 5.2 or revision			
CMETS + JCC Layer 4	GNA/ST II Application ID, LTA Application ID, Application ID under Enhancement 5.2 or revision	JCC connectivity_applicant free text	Split all IDs from CMETS cells. Search each ID inside JCC connectivity_applicant, in GNA -> LTA -> 5.2 order.	TGNA, GNA, Source
JCC bay number extraction	Matched CMETS row from Layer 4	JCC schedule_current_jcc_ists_scope / ists_scope	Once the CMETS row is identified in JCC by GNA -> LTA -> 5.2 ID, read the "Under ISTS Scope Connectivity / Transmission System" text. If a bay number is present, use it as the bay number.	Bay No (J Bay No (E Allocation Module 6
CMETS + Bay Allocation fallback	Voltage level, Name of Developers	Bay lookup voltage bucket and entity_name	Run only when the matched JCC row does not provide a bay number. Normalize voltage to 220kv or 400kv; then match developer/entity by exact, substring, or parenthetical core-name comparison.	Bay No (E Allocation Substation (Bay Allocation Substation Coordination Allocation Source

CMETS Columns

Column	Source	Rule in plain English
PDF	CMETS internal	File path/name of the source PDF.
Page Number	CMETS internal	Page number where the row was extracted.
CMETS GNA Approved	CMETS meeting metadata	Meeting number is placed here when the PDF is classified as GNA. Meeting number patterns include 42nd Consulting Meeting, 42nd CMETS, CMETS_NR/42, or Meeting No. 42.
CMETS LTA Approved	CMETS meeting metadata	Meeting number is placed here when LTA keywords are clearly dominant.
CMETS GNA Meeting Date	CMETS meeting metadata	Date from first page, or page 2 fallback, written as dd.mm.yyyy when PDF is GNA. Pattern examples: 11th November 2025, 25 April 2025, November 11, 2025.
CMETS LTA Meeting Date	CMETS meeting metadata	Same date logic, but filled when PDF is LTA.

Column	Source	Rule in plain English
Substation	CMETS PDF text	Extracted from headers like Connectivity Location, Nearest Pooling Station, Connectivity Granted at, Connectivity Injection Point, Substation. Value patterns include names like Bhadla-V or Aligarh (PG) .
Project Location	CMETS PDF text	Extracted directly when Project Location appears.
Name of Developers	CMETS PDF text	Extracted from Applicant/Developer columns. Company-name patterns help identify it. Values containing LOA/criterion/applying artefacts are rejected.
GNA/ST II Application ID	CMETS PDF text	GNA or Stage-II application ID. Accepted values are normalized as numeric IDs and kept as one of the primary keys for Effectiveness and JCC matching.

- Remove duplicate rows using GNA, LTA, Project Location, and Developer as the key.

Effectiveness Columns

Column	Source	Rule in plain English
source_file	Effectiveness internal	Source effectiveness PDF name.
sl_no	Effectiveness PDF	Serial number from the table.
application_id	Effectiveness PDF	Main ID used to build the lookup for mapping.
name_of_applicant	Effectiveness PDF	Applicant name. Can overwrite CMETS developer name when IDs match.
region	Effectiveness PDF	Region. Populates mapped Region .
type_of_project	Effectiveness PDF	Project type such as Solar, Wind, Hybrid, Hydro, ESS. Populates mapped Type of Project .
installed_capacity_mw	Effectiveness PDF	Numeric float. Can overwrite CMETS Application Quantum.
solar_mw	Effectiveness PDF	Numeric solar capacity. Populates Installed capacity (MW) solar .
wind_mw	Effectiveness PDF	Numeric wind capacity. Populates Installed capacity (MW) wind .
ess_mw	Effectiveness PDF	Numeric ESS/BESS capacity. Populates Installed capacity (MW) ess .
hydro_mw	Effectiveness PDF	Numeric hydro/pump-storage capacity. Populates Installed capacity (MW) hydro .
connectivity_mw	Effectiveness PDF	Numeric connectivity capacity. Listed as an overlapping update candidate but current merge updates Application Quantum using installed capacity first.
present_connectivity_mw	Effectiveness PDF	Current/deemed GNA connectivity. Extracted and exported, not directly mapped into CMETS columns in current merge.

Column	Source	Rule in plain English
substation	Effectiveness PDF	Can overwrite CMETS Substation when IDs match.
state	Effectiveness PDF	Can overwrite CMETS State when IDs match.
expected_date	Effectiveness PDF	Used to update GNA Operationalization Date and Additional Capacity Date. Date parsing accepts numeric and month-name formats.

Effectiveness Mapping Rules

- Build lookup as `application_id -> effectiveness record`.
- Disk JSON cache overrides the in-memory current extraction if the same application ID appears.
- On a CMETS match, update only when the effectiveness value is not blank/null/NA.
- Date rule: if CMETS date is empty, use effectiveness date. If effectiveness date is later, update. If CMETS date is same/later, keep CMETS date.

Effectiveness-Derived Mapped Columns

Output column	Source	Rule
Region	Effectiveness region	Filled on ID match.
Type of Project	Effectiveness type_of_project	Filled on ID match.
Installed capacity (MW) solar	Effectiveness solar_mw	Filled on ID match.
Installed capacity (MW) wind	Effectiveness wind_mw	Filled on ID match.
Installed capacity (MW) ess	Effectiveness ess_mw	Filled on ID match.
Installed capacity (MW) hydro	Effectiveness hydro_mw	Filled on ID match.
Installed capacity (MW) hybrid	Effectiveness component MW	If effectiveness type is hybrid, sum solar + wind + ESS + hydro.
Installed/Break-up Capacity (MW) Solar	Effectiveness + CMETS Type	Sum effectiveness solar MW and CMETS Type solar MW.
Installed/Break-up Capacity (MW) Wind	Effectiveness + CMETS Type	Sum effectiveness wind MW and CMETS Type wind MW.
Installed/Break-up Capacity (MW) Hybrid	Effectiveness + CMETS Type	If project is hybrid or has multiple categories, sum all effectiveness MW plus all CMETS Type MW.
Installed/Break-up Capacity (MW) Hydro	Effectiveness + CMETS Type	Sum effectiveness hydro MW and CMETS hydro/PSP type MW.

JCC Columns

Column	Source	Rule in plain English
source_pdf	JCC internal	Source JCC PDF name.
page_number	JCC internal	Page where the table row was found.
sr_no	JCC PDF table	Serial number.
pooling_station	JCC PDF table	Exact raw text from Pooling Station .
connectivity_applicant	JCC PDF table	Exact raw text from Connectivity Applicant . This is where CMETS GNA/LTA/5.2 IDs are searched.
connectivity_quantum_mw	JCC PDF table	Exact raw text from Connectivity Quantum (MW) .
schedule_as_per_current_jcc	JCC PDF table	Exact raw text from Under Grantee scope Gen Commissioning /Connectivity line schedule . Used later to compute total COD and COD_Found .
schedule_current_jcc_ists_scope	JCC PDF table	Text from "Under ISTS Scope Connectivity / Transmission System". It may contain bay number details. It is not used for GNA/TGNA MW calculation, but after a CMETS row is matched to this JCC row, the code extracts bay number from this text first.
connectivity_start_date_under_gna	JCC PDF table	Exact raw text from Connectivity Start Date under GNA and Connectivity Effectiveness date . Used later to extract effective_date .
total COD	Computed	Sum of MW values found under the Generation: section only, when those MW entries have a date and mention COD or Commissioned .
COD_Found	Computed	True when qualifying Generation: entries with MW, date, and COD/Commissioned are found. Otherwise False .
effective_date	Computed	Date extracted from connectivity_start_date_under_gna , for example 12.12.2025 from text like Connectivity effective w.e.f. 12.12.2025 .
TGNA	Computed	Filled with total COD only when COD_Found is True and effective_date is after the current run date. Otherwise blank.
GNA	Computed	Filled with total COD only when COD_Found is True and the current run date is after effective_date . Otherwise blank.
remarks	JCC PDF table	Remarks from JCC table.

JCC Extraction Rules

- A JCC page must contain **Pooling**, **Quantum**, and **Connectivity**.
- A table is accepted when it contains the target JCC columns: **Pooling Station**, **Connectivity Applicant**, **Connectivity Quantum (MW)**, **Under Grantee scope Gen Commissioning /Connectivity line schedule**, and **Connectivity Start Date under GNA and Connectivity Effectiveness date**.
- The JCC extractor and handler should use the LLM for extraction.
- Raw JCC extraction must be exact column extraction from the source table. Do not use logical inference while filling the raw columns.
- Header rows are skipped when they contain pooling/grantee scope/under ISTS wording.

JCC Mapping and Computed Columns

Output column	Source	Rule
total COD	Matched JCC row	Read schedule_as_per_current_jcc . Under Generation: only, sum MW entries that also have a date and mention COD or Commissioned .
COD_Found	Matched JCC row	True when at least one qualifying Generation: COD/Commissioned MW entry is found; otherwise False .
effective_date	Matched JCC row	Extract only the date from connectivity_start_date_under_gna , such as 12.12.2025 .
TGNA	Matched JCC row	If COD_Found is True and effective_date is after the current run date, write total COD ; otherwise keep blank.
GNA	Matched JCC row	If COD_Found is True and the current run date is after effective_date , write total COD ; otherwise keep blank.
Match Source	Mapping logic	Shows whether row matched through GNA, LTA, or 5.2 ID.
Bay No (JCC)	Matched JCC row	Extracted from schedule_current_jcc_ists_scope / ists_scope after the CMETS row is matched to JCC. Multiple bay numbers are joined with .

Layer 4 now runs every pipeline execution. For every CMETS row, it searches JCC **connectivity_applicant** in this order: GNA ID, then LTA ID, then 5.2 ID. Once the JCC row is identified, the same matched row is used for GNA/TGNA and for extracting the JCC bay number from the ISTS-scope column.

JCC COD, TGNA, and GNA Logic

The JCC handler should follow this logic in **pipeline/jcc_handler/jcc_output_layer.py**.

1. Matching happens first. CMETS IDs are searched inside JCC **connectivity_applicant**. The ID cascade is **GNA/ST II Application ID -> LTA Application ID -> Application ID under Enhancement 5.2 or revision**.
2. After a JCC row is matched, read **schedule_as_per_current_jcc**, which is the raw text from **Under Grantee scope Gen Commissioning /Connectivity line schedule**.
3. Look only inside the **Generation:** section. Ignore line schedule, transmission, ISTS, or any other non-generation section.
4. Add MW values only when the same entry has a date and mentions **COD** or **Commissioned**.
5. Write the sum to **total COD** and set **COD_Found = True**. If no qualifying entry is found, set **COD_Found = False** and keep **total COD**, **effective_date**, **TGNA**, and **GNA** blank.
6. If **COD_Found = True**, extract **effective_date** from **connectivity_start_date_under_gna**. For example, from **Connectivity effective w.e.f. 12.12.2025**, keep only **12.12.2025**.

- 7. Compare `effective_date` with the current run date. If the current run date is after `effective_date`, write `total COD` to `GNA`.
- 8. If `effective_date` is after the current run date, write `total COD` to `TGNA`.
- 9. `GNA` and `TGNA` are mutually exclusive for a matched JCC row. If `COD_Found = False`, both stay blank.

Example generation schedule text:

```
Generation:
110 MW: 12.06.2023 (Commissioned)
50 MW: 05.06.2023 (Commissioned)
30 MW: 10.06.2023 (Commissioned)
```

Here `total COD = 110 + 50 + 30 = 190` and `COD_Found = True`.

```
Generation:
110 MW: 20.06.2023 (COD)
50 MW: 05.06.2023 (COD)
30 MW: 10.06.2023 (COD)
```

This also gives `total COD = 190` and `COD_Found = True`. If the entries do not mention `COD` or `Commissioned`, `COD_Found = False` and `TGNA/GNA` remain blank.

Bay Allocation Columns

Full PDF Table Columns

Column	Source	Rule in plain English
<code>sl_no</code>	Bay Allocation table col 0	Starts a new substation when numeric.
<code>name_of_substation</code>	Col 1	Substation name.
<code>substation_coordinates</code>	Col 2	Coordinates of the substation.
<code>region</code>	Col 3	Region.
<code>transformation_capacity_planned_mva</code>	Col 4	Planned transformation capacity.
<code>transformation_capacity_existing_mva</code>	Col 5	Existing transformation capacity.
<code>transformation_capacity_under_implementation_mva</code>	Col 6	Under-implementation transformation capacity.
<code>bay_no_220kv</code>	Col 7	RE capacity granted 220kV bay number.
<code>connectivity_quantum_mw_220kv</code>	Col 8	220kV connectivity quantum.
<code>name_of_entity_220kv</code>	Col 9	220kV entity/developer name.

Column	Source	Rule in plain English
bay_no_400kv	Col 10	RE capacity granted 400kV bay number.
connectivity_quantum_mw_400kv	Col 11	400kV connectivity quantum.
name_of_entity_400kv	Col 12	400kV entity/developer name.
margin_bay_no_220kv	Col 13	220kV margin bay number.
margin_available_mw_220kv	Col 14	220kV margin available.
margin_bay_no_400kv	Col 15	400kV margin bay number.
margin_available_mw_400kv	Col 16	400kV margin available.
space_provision_220kv	Col 17	Space provision for 220kV line bays.
space_provision_400kv	Col 18	Space provision for 400kV line bays.
remarks	Col 19	Remarks.

Bay Allocation Excel Columns

Column	Source	Rule
source_pdf	Internal	Bay Allocation PDF name.
page_number	Internal	Page number.
sl_no	Extracted	Substation serial number.
name_of_substation	Extracted	Substation name.
substation_coordinates	Extracted	Coordinates.
region	Extracted	Region.
220kv_bay_no	Extracted dict	Serialized as bay: entity bay: entity.
400kv_bay_no	Extracted dict	Serialized as bay: entity bay: entity.

Bay Mapping Columns

Output column	Source	Rule
Bay No (JCC)	Layer 4 JCC match	Extracted from the matched JCC row's "Under ISTS Scope Connectivity / Transmission System" text. This is the preferred bay source.
Bay No (Bay Allocation)	JCC first, else Bay Allocation lookup	If Bay No (JCC) is present, copy it here and skip Bay Allocation search for that row. If JCC has no bay number, match CMETS voltage to 220kV/400kV and match CMETS developer name to bay entity name by exact/substring/core-name comparison. Multiple bay numbers are joined by .
Substation Name (Bay Allocation)	Bay Allocation	From matched bay allocation substation. Duplicates are removed. Populated only when the Bay Allocation fallback is

Output column	Source	Rule
Allocation)	lookup	used.
Substation Coordinates (Bay Allocation)	Bay Allocation lookup	From matched bay allocation substation. Populated only when the Bay Allocation fallback is used. If coordinates already exist and no JCC bay number is present, the row is skipped and not searched again.
Bay No Source	Mapping logic	JCC Under ISTS Scope when the bay number came from JCC; Bay Allocation PDF when it came from the fallback lookup.

Bay Number Selection Logic

1. Module 4 Layer 4 matches each CMETS row to a JCC row by searching CMETS IDs inside JCC `connectivity_applicant` in GNA -> LTA -> 5.2 order.
2. After the JCC row is identified, the code reads `schedule_current_jcc_ists_scope / ists_scope`, which corresponds to "Under ISTS Scope Connectivity / Transmission System".
3. If that text contains a bay number, Bay No (JCC) is populated and Module 6 copies it into Bay No (Bay Allocation) with Bay No Source = JCC Under ISTS Scope.
4. If the matched JCC row does not contain a bay number, Module 6 searches Bay Allocation PDFs for that row using voltage plus developer/entity name.
5. If Bay Allocation fallback matches, Module 6 writes bay number, substation name, coordinates, and Bay No Source = Bay Allocation PDF.

Exact Row Identification and Source Dependency

The pipeline starts from a CMETS row. Other sources do not create the final row by themselves; they enrich the already-selected CMETS row.

Source used to enrich CMETS row	Exact CMETS column names used to identify/search	Exact source column names searched	How the source row is identified
Effectiveness PDF	GNA/ST II Application ID, then LTA Application ID, then Application ID under Enhancement 5.2 or revision	application_id	Split IDs from the CMETS columns. Search them in <code>application_id</code> in priority order. First matching effectiveness record wins.
JCC PDF	GNA/ST II Application ID, then LTA Application ID, then Application ID under Enhancement 5.2 or revision	connectivity_applicant	Split IDs from the CMETS columns. Search each ID inside JCC <code>connectivity_applicant</code> in priority order. First matching JCC row wins.

Source used to enrich CMETS row	Exact CMETS column names used to identify/search	Exact source column names searched	How the source row is identified
Bay Allocation PDF fallback	Voltage level, Name of Developers	Voltage bucket from 220kv.bay_no / 400kv.bay_no; entity text from Bay Allocation name_of_entity_220kv or name_of_entity_400kv	Normalize CMETS Voltage level to 220kv or 400kv. Search only that voltage bucket. Match CMETS Name of Developers against Bay Allocation entity name by exact, substring, or core-name comparison.

Bay Allocation does not depend on CMETS application IDs. It depends on CMETS Voltage level and Name of Developers, but only after the JCC bay-number check fails.

Exact Column Lineage for Bay Mapping

Final output column	Preferred source	Fallback source	Exact source columns used	Rule
Bay No (JCC)	JCC PDF	None	JCC schedule_current_jcc_ists_scope / ists_scope	After the JCC row is identified through CMETS IDs, extract bay number from the "Under ISTS Scope Connectivity / Transmission System" text.
Bay No (Bay Allocation)	JCC Bay No (JCC)	Bay Allocation PDF	JCC schedule_current_jcc_ists_scope / ists_scope; fallback Bay Allocation 220kv.bay_no or 400kv.bay_no	If JCC bay exists, copy it here. If not, use Bay Allocation fallback matching and write matched bay number(s).
Substation Name (Bay Allocation)	Bay Allocation PDF	None	Bay Allocation name_of_substation	Populated only when Bay Allocation fallback finds a matched row.
Substation Coordinates (Bay Allocation)	Bay Allocation PDF	None	Bay Allocation substation_coordinates	Populated only when Bay Allocation fallback finds a matched row.
Bay No Source	Mapping logic	Mapping logic	Derived from whether JCC or Bay Allocation supplied the bay number	JCC Under ISTS Scope when JCC supplied the bay number;

Final output column	Preferred source	Fallback source	Exact source columns used	Rule
				Bay Allocation PDF when fallback supplied it.

Bay Allocation PDF Columns Extracted

The Bay Allocation PDF extraction stores each substation as one record with 220kV and 400kV bay dictionaries.

Extracted field in JSON / Excel	Exact PDF table column	How it is used later
name_of_substation	Name of Substation	Written to Substation Name (Bay Allocation) after fallback match.
substation_coordinates	Coordinates / substation coordinate column	Written to Substation Coordinates (Bay Allocation) after fallback match.
region	Region	Stored in lookup entry; not currently written to bay output columns.
220kv.bay_no / 220kv_bay_no	220kV Bay No. plus 220kV Name of Entity	Builds lookup entries for CMETS rows where Voltage level normalizes to 220kv.
400kv.bay_no / 400kv_bay_no	400kV Bay No. plus 400kV Name of Entity	Builds lookup entries for CMETS rows where Voltage level normalizes to 400kv.

Bay Allocation Fallback Row Match in Detail

1. Read CMETS Voltage level.
2. Normalize Voltage level:
 - values containing 220 become 220kv
 - values containing 400 become 400kv
 - any other value is skipped as no_voltage
3. Read CMETS Name of Developers.
4. Pick Bay Allocation lookup entries only from the matching voltage bucket.
5. Compare CMETS Name of Developers with Bay Allocation entity name from the selected bucket:
 - exact normalized name match
 - CMETS name contained in Bay Allocation entity name
 - Bay Allocation entity name contained in CMETS name
 - core-name match after removing parenthetical capacity text
6. If one or more entries match, deduplicate by bay number and write:
 - matched bay numbers to Bay No (Bay Allocation)
 - matched name_of_substation to Substation Name (Bay Allocation)
 - matched substation_coordinates to Substation Coordinates (Bay Allocation)
 - Bay No Source = Bay Allocation PDF